

Advanced Computer Algorithms (ET6501701)

Place: TBA

Fall 2025

Instructor: M. B. Lin

Course objective: The objective of this course is to familiarize the student with the process of implementing a system-level algorithm at the RTL level in Verilog HDL. Upon completion, the student will be able to carry out the RTL (register-transfer level) design and implementation of a digital IC module (or IP) from a system-level algorithm. To reach this, this course will also focus on the applications of pipeline and parallel techniques to the RTL design and implementation of system-level algorithms.

Prerequisites: You must have read the following textbooks or their equivalent.

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, *Introduction to Algorithms*, 4th ed., The MIT Press and McGraw-Hill Book Company, New York, 2022 (Chapters 1 to 16.) (Note that **any version** is fine.)
2. Ming-Bo Lin, *FPGA Class Notes*, 2025.
3. Ming-Bo Lin, *Digital System Designs and Practices: Using Verilog HDL and FPGAs*, John Wiley & Sons, Inc., 2008. (ISBN-13: 978-0470823231)

Textbooks:

1. *Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, *Introduction to Algorithms*, 4th ed., The MIT Press and McGraw-Hill Book Company, New York, 2022. (Note that **any version** is fine.)
2. Behrouz A. Forouzan, *Introduction to Cryptography and Network Security*, New York: McGraw-Hill, 2008.
3. William Stallings, *Cryptography and Network Security Principles and Practice*, 8th Global ed., Pearson Education Limited, 2023.

References:

1. Robert Sedgewick and Kevin Wayne, *Algorithms*, 4th ed., Upper Saddle River, NJ: Addison-Wesley, 2011.
2. Udi Manber, *Introduction to Algorithms — A Creative Approach*, Reading, Massachusetts: Addison-Wesley, 1989.
3. Ellis Horowitz, Sartaj Sahni, and Sanguthevar Rajasekaran, *Computer Algorithms*, New York: Computer Science Press, 1997.
4. Mark Allen Weiss, *Data Structures & Algorithm Analysis in C++*, 4th ed., Upper Saddle River, NJ: Addison-Wesley, 2014.
5. Zvi Kohavi and Niraj K. Jha, *Switching and Finite Automata Theory*, 3rd ed., Cambridge University Press, 2010.

Course Outline:

1. **Parallel-prefix sums:** Introduction to parallel algorithms, introduction to prefix sum, parallel-prefix sum architecture, parallel-prefix adders, and applications
2. **Sorting networks.** (Chapter 27, Second edition): Comparison networks, the zero-one principle, a bitonic sorting network, a merging network, and a sorting network.
 - Hardware priority queue
 - Iterative Implementation
 - Parallel Implementation
3. **Number-theoretic algorithms.** (Chapter 31): Modular arithmetic, solving modular linear equations, the Chinese remainder theorem, powers of an element, the RSA public-key cryptosystem, primality testing, and integer factorization.
4. **RSA cryptosystem.** RSA encryption and decryption, modular multiplication, two-adder implementation of RSA cryptography, Montgomery implementation of RSA cryptography, and Karatsuba multiplier.
5. **Finite fields and advanced encryption standard.** (Forouzan, Chapters 4 and 7): Finite fields of the form $GF(2^8)$, polynomial arithmetic, finite field arithmetic, advanced encryption standard (AES), and Reed-Solomon encoders/decoders.
 - The AES algorithm
 - Implementation of the AES algorithm
 - Reed-Solomon encoders/decoders
6. **Polynomials and the FFT.** (Chapter 30): Representation of polynomials, the DFT and FFT, efficient FFT implementation.
 - Implementation of the CORDIC algorithm
 - FFT implementation with the CORDIC algorithm
7. **NP-Completeness.** (Chapter 34): Polynomial time, polynomial-time verification, NP-completeness and reducibility, NP-completeness proofs, and NP-complete problems.
8. **Approximation algorithms.** (Chapter 35): The vertex-cover problem, the traveling-salesman problem, the set-covering problem, and the subset-set problem.

Homeworks: One simple term project.

Grading method: Homework (one simple term project): 40%; mid exam: 30%; final exam: 30%; bonus: class participation and extra homework (if any).

Nothing great was ever achieved without enthusiasm.

R. W. Emerson.

The reward of a thing well done is to have done it.

R. W. Emerson.

The only limit to our realization of tomorrow will be our doubts of today.

Franklin D.

Roosevelt.
